

CO5_ASSESSMENT 1

CONSTRAINT -BASED PROBLEM SOLVING

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Course : Computer Vision

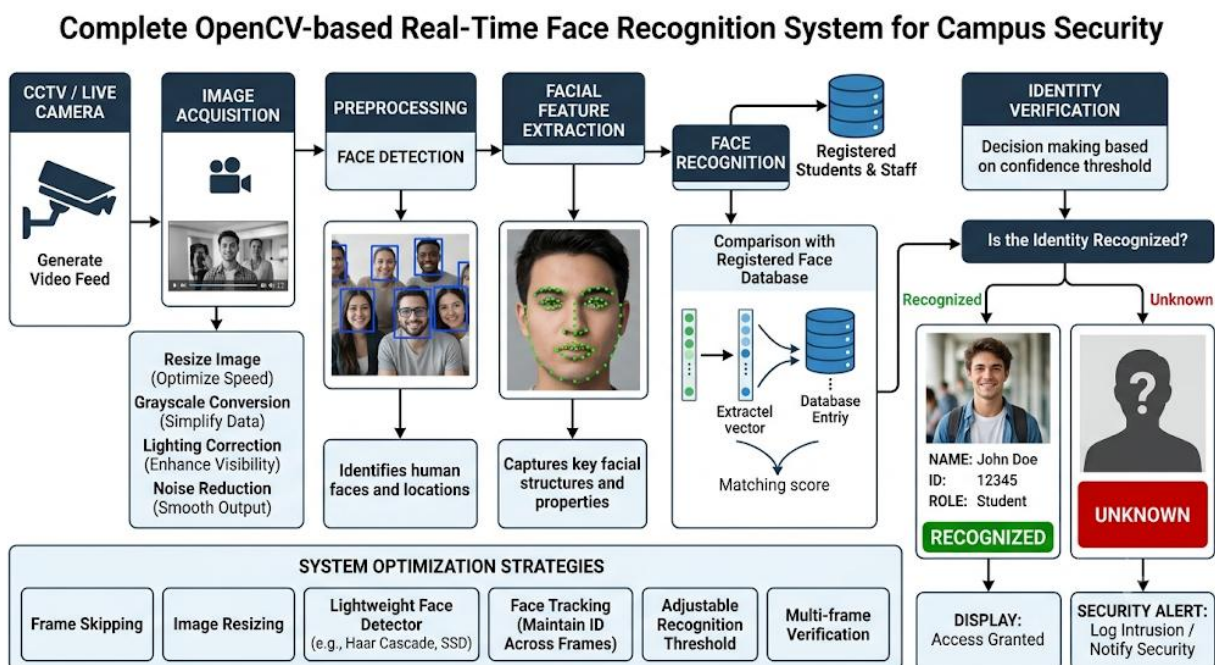
1. Real-Time Face Recognition for Campus Security

Problem Statement

A university wants to implement a camera-based security system that can identify registered students and staff in real time. The system must work under different lighting conditions, minimize false recognition, and maintain real-time performance on a low-cost computer.

Proposed OpenCV Pipeline

Image Acquisition → Preprocessing → Face Detection → Feature Extraction → Face Recognition → Verification → Identification → Display/Alert



1. Image Acquisition

The system uses an OpenCV camera interface such as `cv2.VideoCapture()` to continuously capture frames from the CCTV/video stream.

Each frame is processed individually to detect and recognize faces.

2. Preprocessing

The captured frame is first converted into a suitable format.

Steps include:

- Resize the frame to reduce processing time.
- Convert the image to grayscale.
- Apply histogram equalization or CLAHE to improve visibility under different lighting conditions.
- Apply a small Gaussian blur to reduce image noise.

3. Face Detection

OpenCV Haar Cascade or another lightweight face detector can be used to locate faces.

The detector returns bounding boxes around detected faces. Only the detected face regions are passed to the next stage.

4. Feature Extraction

The detected face is normalized by resizing it to a fixed size.

Facial features such as the shape and structure of the face are extracted using a suitable face-recognition technique. The extracted features are converted into a feature representation that can be compared with registered individuals.

5. Face Recognition

During registration, facial features of authorized students and staff are stored in a database.

For every detected face, the extracted features are compared with the stored features. If the similarity exceeds a predefined threshold, the person is identified as a registered individual.

If the similarity is below the threshold, the person is classified as Unknown.

6. False Recognition Reduction

False recognition can be minimized by:

- Using a suitable recognition threshold.
- Capturing multiple samples of each registered person.
- Using good-quality face images during registration.
- Rejecting very small or unclear faces.
- Confirming recognition across multiple consecutive frames.

7. Real-Time Optimization

Since the system must run on a low-cost computer, the following optimizations can be applied:

- Reduce input frame resolution.
- Process every second or third frame instead of every frame.
- Detect faces in a resized frame.
- Use lightweight face detection.
- Perform recognition only when a face is detected.
- Track already detected faces instead of detecting them repeatedly.

8. Output

The system displays:

- Bounding box around each face.
- Recognized person's name.
- Recognition confidence.
- Unknown-person warning when no registered match is found.

Conclusion

The proposed OpenCV pipeline combines preprocessing, face detection, feature extraction, recognition, and optimization. Lightweight detection, reduced image resolution, frame skipping, and tracking help maintain real-time performance on low-cost hardware. Multiple-frame verification and suitable recognition thresholds reduce false recognition. Therefore, the system can provide an efficient real-time campus security solution.

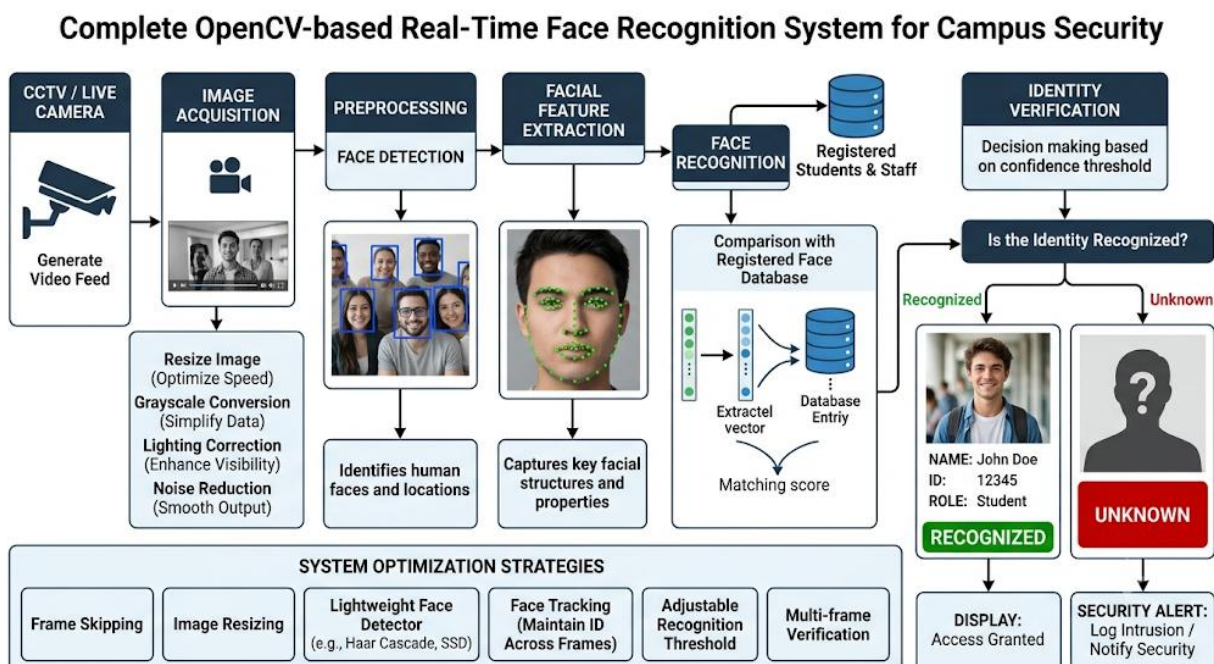
2. Intelligent Traffic Violation Detection System

Problem Statement

The traffic department wants to automatically monitor vehicles using roadside cameras. The system must detect and track vehicles, identify vehicles crossing a predefined line, count them accurately, detect abnormal movement, and display alerts in real time.

Proposed OpenCV Pipeline

Video Acquisition → Preprocessing → Vehicle Detection → Tracking → Line Crossing
Detection → Counting → Abnormal Movement Detection → Visualization/Alert



1. Video Acquisition

The system captures live video using an OpenCV camera interface.

Each frame is processed continuously to identify and track vehicles.

2. Preprocessing

Preprocessing improves the quality of video frames.

The following operations can be used:

- Resize frames for faster processing.
- Convert frames to grayscale when required.
- Apply Gaussian blur to reduce camera noise.
- Apply suitable filtering to reduce unwanted variations.

3. Vehicle Detection

Vehicles can be detected using background subtraction or an object-detection method.

For a fixed roadside camera, background subtraction can identify moving regions. Contours can then be extracted from the foreground mask to locate vehicles.

4. Vehicle Tracking

Each detected vehicle is assigned a unique ID.

The position of each vehicle is tracked across consecutive frames using its bounding box or centroid.

This prevents the same vehicle from being counted repeatedly.

5. Line-Crossing Detection

A virtual line is drawn across the road.

The centroid of each vehicle is monitored. When a vehicle centroid moves from one side of the line to the other, the system identifies it as a line-crossing event.

The vehicle count is then increased by one.

6. Abnormal Movement Detection

The system can analyze the direction and speed of tracked vehicles.

Possible abnormal movements include:

- Sudden stopping.
- Moving in the wrong direction.
- Unusual changes in trajectory.
- Movement outside the expected road region.

An alert is generated when suspicious movement is detected.

7. Handling Shadows and Noise

Shadows can cause false vehicle detection. To reduce this problem:

- Use shadow-removal techniques.
- Apply morphological operations.
- Use suitable foreground masks.
- Ignore very small contours.

Camera noise can be reduced using Gaussian or median filtering.

8. Dynamic Background

Trees, changing illumination, and other background objects can create false detections.

This can be reduced using:

- Adaptive background models.
- Morphological opening and closing.

- Minimum contour-area thresholds.
- Region-of-interest restrictions.

9. Multiple Vehicle Handling

Each detected vehicle receives a unique tracking ID. The system maintains the position of every vehicle across frames.

Therefore, multiple vehicles can be tracked simultaneously without repeatedly counting the same vehicle.

10. Visualization

The system displays:

- Vehicle bounding boxes.
- Vehicle tracking IDs.
- Virtual counting line.
- Total vehicle count.
- Abnormal movement alerts.

Conclusion

The proposed OpenCV system combines video acquisition, preprocessing, vehicle detection, tracking, line-crossing analysis, and visualization. Filtering, morphological operations, region restrictions, and tracking IDs help reduce false detections and duplicate counting. Therefore, the system can provide accurate real-time traffic monitoring.

3. Automated Document Processing System

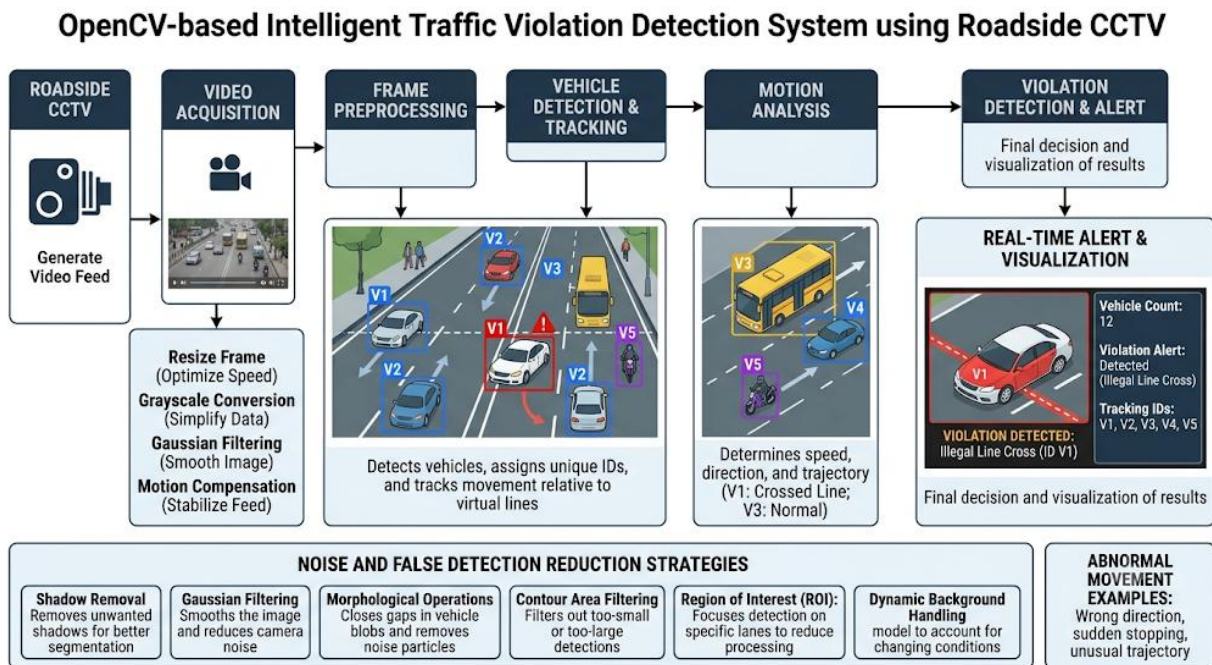
Problem Statement

A company receives thousands of scanned documents containing blurred text, uneven illumination, noise, and different font sizes. The system must improve document quality, remove

noise and background, prepare images for OCR, segment text regions, and process documents efficiently in batch mode.

Proposed OpenCV Pipeline

Input Document → Grayscale Conversion → Noise Filtering → Illumination Correction → Thresholding → Morphological Operations → Segmentation → Geometric Correction → **OCR**



1. Image Acquisition

The scanned document is loaded into OpenCV using an image-reading function.

The system can process individual images or a large collection of documents in batch mode.

2. Grayscale Conversion

The document is converted from RGB/color format to grayscale.

Grayscale conversion reduces the image to a single intensity channel and simplifies further processing.

3. Filtering

Noise is removed using filters such as:

- Gaussian filter.
- Median filter.

Filtering removes small unwanted variations and improves the quality of characters.

4. Illumination Correction

Uneven lighting can make some parts of the document darker or brighter.

Brightness normalization or adaptive thresholding can be used to handle uneven illumination and improve text visibility.

5. Thresholding

Thresholding converts the grayscale document into a binary image.

Adaptive thresholding is useful when illumination is not uniform because it determines the threshold locally.

This separates text from the background.

6. Morphological Operations

Morphological operations such as erosion and dilation can improve the binary document.

Opening can remove small noise.

Closing can connect broken characters and small gaps.

These operations improve the structure of text regions before OCR.

7. Text Segmentation

Contours and connected components can be used to identify text regions.

The document can be divided into lines, words, or blocks so that OCR can process the relevant regions efficiently.

8. Geometric Correction

Scanned documents may be rotated or tilted.

Perspective transformation and image rotation can correct the document orientation and make text horizontal.

This improves OCR recognition.

9. OCR Preparation

After preprocessing, the final document image should have:

- Clear text.
- Reduced noise.
- Uniform illumination.
- Correct orientation.
- Suitable contrast.

The processed image is then passed to an OCR engine.

10. Batch Processing

For thousands of documents, the system processes images automatically in batches.

This reduces manual work and provides consistent preprocessing for every document.

Conclusion

The OpenCV pipeline improves OCR accuracy by sequentially removing noise, correcting illumination, separating text from the background, improving character structure, segmenting text, and correcting document geometry. Batch processing makes the system suitable for large-scale document processing.

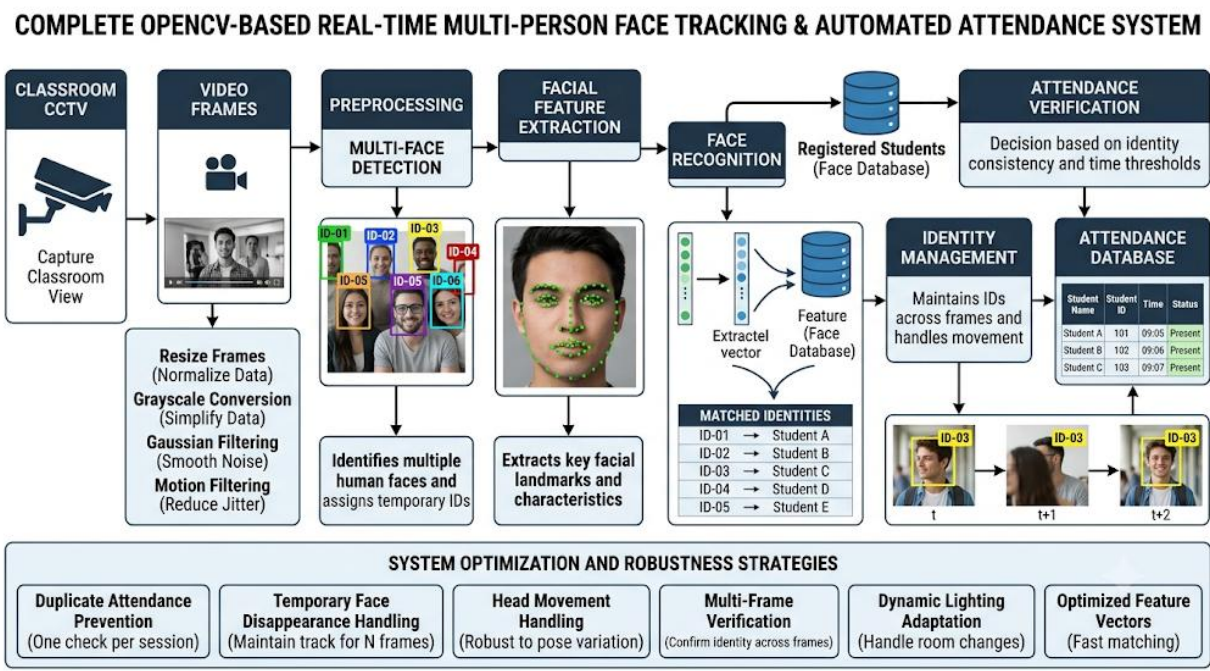
4. Real-Time Multi-Person Tracking and Attendance System

Problem Statement

A college wants to automatically record student attendance using classroom CCTV cameras. The system must detect multiple faces, track people across frames, recognize registered students, avoid duplicate attendance, handle temporary face disappearance, and operate with minimum processing delay.

Proposed Pipeline

CCTV Video → Frame Preprocessing → Multi-Face Detection → Feature Extraction → Face Recognition → Tracking → Identity Management → Attendance Verification → Record Attendance



1. Video Acquisition

The classroom CCTV camera continuously provides video frames to the OpenCV system.

2. Preprocessing

Each frame is resized to reduce computation.

Grayscale conversion and suitable filtering can be applied to improve face detection and reduce noise.

3. Multi-Face Detection

A face detector identifies all visible faces in the frame.

Each detected face receives a bounding box.

The system can process multiple detected faces simultaneously.

4. Feature Extraction

For every detected face, facial features are extracted and converted into a feature representation.

These features are compared with the registered student database.

5. Face Recognition

The system compares the extracted features with stored student features.

If the similarity is above the recognition threshold, the student is identified.

Otherwise, the face is marked as unknown.

6. Tracking

Each detected person is assigned a unique tracking ID.

The system tracks the person's position across consecutive frames.

Even when a face is temporarily not detected, the tracking system can maintain the person's identity for a short period.

7. Attendance Management

When a registered student is confidently recognized, the system checks whether the student has already been marked present.

A record is created only if the student is not already marked present.

Example:

Student ID → Name → Date → Time → Attendance Status

8. Preventing Duplicate Attendance

Duplicate attendance is prevented by maintaining a list or database of students already marked present for the current class session.

If the student's ID already exists in the attendance record, another entry is not created.

9. Handling Head Movement

The system can use multiple frames for recognition instead of depending on a single frame.

Tracking also helps maintain identity when the student's face changes slightly due to head movement.

10. Temporary Face Disappearance

If a student turns away or is temporarily blocked, the tracking system maintains the previous tracking ID for a short duration.

When the face becomes visible again, the system can reconnect the detection with the existing identity.

11. Real-Time Optimization

To reduce processing delay:

- Resize input frames.
- Process selected frames.
- Use lightweight face detection.
- Track detected people instead of repeatedly detecting everything.
- Perform recognition only when required.

Conclusion

The proposed system combines multi-face detection, feature extraction, recognition, and tracking to maintain student identities across frames. A unique identity and attendance record prevent duplicate entries. Tracking also handles temporary face disappearance and slight head movements. Therefore, the system can provide continuous and efficient automated attendance.

5. Smart Home Surveillance and Intrusion Detection

Problem Statement

A smart home security system uses a fixed camera to continuously monitor an entrance. It must detect significant motion, distinguish real movement from camera noise and background changes, track moving objects, identify suspicious activity, and generate alerts while using limited computational resources.

Proposed OpenCV Pipeline

Video Acquisition → Background Subtraction → Filtering → Morphological Operations → Contour Detection → Motion Analysis → Object Tracking → Region Analysis → Intrusion Alert

1. Video Acquisition

A fixed camera continuously captures video using OpenCV.

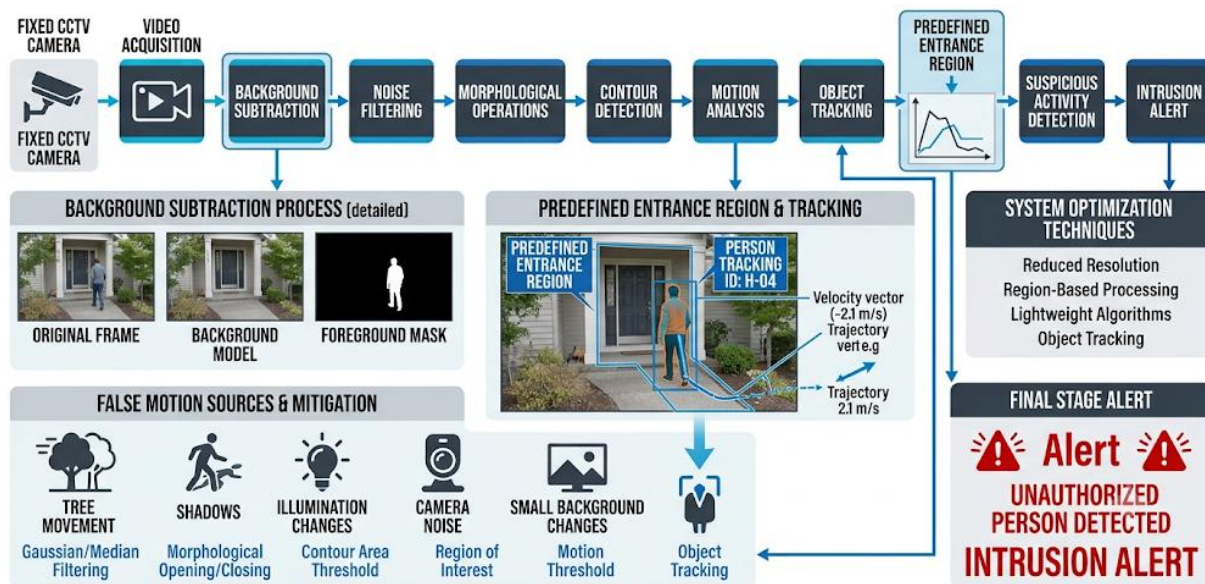
The frames are processed sequentially to detect movement around the entrance.

2. Background Subtraction

Background subtraction separates moving objects from the static background.

The background model represents the normal appearance of the entrance. Significant changes are identified as foreground regions.

SMART HOME SURVEILLANCE & INTRUSION DETECTION: AN OPENCV-BASED SYSTEM PIPELINE



3. Filtering

Noise in the camera frames can create false motion.

Gaussian or median filtering can reduce small random variations before motion detection.

4. Morphological Operations

Morphological operations improve the foreground mask.

- **Opening** removes small noise.
- **Closing** fills small gaps in detected objects.
- **Dilation** can connect nearby foreground regions.

5. Contour Detection

Contours are extracted from the foreground mask.

Very small contours are ignored using a minimum area threshold. Larger contours are considered potential moving objects.

6. Motion Analysis

The system analyzes changes in the position of detected objects between consecutive frames.

Small background movements can be ignored by using suitable motion thresholds.

7. Object Tracking

Each moving object is assigned a tracking ID.

The centroid or bounding box of the object is monitored across frames.

This helps determine whether an object is actually moving toward the entrance.

8. Predefined Region

A region of interest is created around the entrance.

If a tracked object enters this predefined region and satisfies the motion conditions, the system treats it as a potential intrusion.

9. Reducing False Alarms

Shadows:

Shadows can be reduced using suitable foreground processing and morphological filtering.

Illumination Changes:

Adaptive background models and suitable thresholds help handle gradual lighting changes.

Trees and Dynamic Backgrounds:

Small or repetitive background movements can be filtered using contour-area thresholds and region-of-interest restrictions.

Camera Noise:

Image smoothing reduces random camera noise.

10. Intrusion Alert

When an unauthorized person or suspicious object enters the predefined region, the system generates an alert.

The alert can contain:

- Time of detection.
- Captured frame.
- Bounding box.
- Tracking ID.
- Intrusion status.

11. Real-Time Optimization**Since computational resources are limited:**

- Use reduced frame resolution.
- Process only required regions.
- Apply lightweight background subtraction.
- Use contour filtering.

Conclusion

The proposed OpenCV surveillance system uses background subtraction, filtering, morphological operations, contour detection, motion analysis, and tracking to detect suspicious activity. Noise, shadows, illumination changes, and dynamic backgrounds are reduced through filtering, thresholds, morphology, and region restrictions. The system therefore provides an efficient real-time intrusion detection solution with limited computational requirements.